Unit 3, Lesson 5: Applying Exponent Laws

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ I can apply the laws of exponents to generate equivalent numerical expressions.

3.5.1 Warm-Up: Math Argument

1. Without performing any calculations, circle the expression you think has the greater value.

60^{63} $\qquad\qquad$ 63^{60}

2. Justify why you selected that expression.

3.5.2 Guided Instruction: Rational Number Bases

1. Consider the table shown.

Exponential Expression	Expanded Form	Equivalent Expression
$\left(\frac{2}{5}\right)^3$	$\frac{2}{5}\cdot\frac{2}{5}\cdot\frac{2}{5}$	$\frac{2^3}{5^3}$
$\left(\frac{2}{5}\right)^2$	$\frac{2}{5}\cdot\frac{2}{5}$	$\frac{2^2}{5^2}$
$\left(\frac{2}{5}\right)^1$	$\frac{2}{5}$	$\frac{2^1}{5^1}$
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$

$\div\frac{2}{5}$

$\div\frac{2}{5}$

$\div\frac{2}{5}$

Complete the statement to describe the pattern in the table.

As the value of the exponent ○ increases ○ decreases by 1 in the column on the left, the equivalent expression in the column on the right can be found by ○ multiplying ○ dividing the previous value by the ○ base. ○ exponent.

Continuing the pattern, the next row in the table would begin with the expression $\left(\frac{2}{5}\right)^{-1}$.

To continue the pattern, find the value of $1\div\frac{2}{5}$. Since dividing by a fraction is equivalent to multiplying by the fraction's reciprocal, the result is as follows.

$$1\div\frac{2}{5} = 1\cdot\frac{5}{2} = \frac{5}{2}$$

Since $\left(\frac{2}{5}\right)^{-1} = \frac{5}{2}$ and the identity exponent law states $\frac{5}{2} = \left(\frac{5}{2}\right)^1$, it can be concluded that $\left(\frac{2}{5}\right)^{-1} = \left(\frac{5}{2}\right)^1$.

2. Continue the pattern.

Exponent	Expanded Form	Equivalent Expression
$\left(\frac{2}{5}\right)^0$	1	$\frac{2^0}{5^0}$
$\left(\frac{2}{5}\right)^{-1}$	$\frac{5}{2}$	$\left(\frac{5}{2}\right)^1$
$\left(\frac{2}{5}\right)^{-2}$		
$\left(\frac{2}{5}\right)^{-3}$		

$\times\frac{5}{2}$

$\times\frac{5}{2}$

$\times\frac{5}{2}$

3. Complete the statement.

A rational base with a negative exponent, $\left(\frac{a}{b}\right)^{-m}$, can be rewritten using the ○ reciprocal ○ opposite of the base with the ○ reciprocal ○ opposite exponent.

4. Rewrite each expression as an equivalent exponential expression with positive exponents.

a. $\left(\frac{2}{3}\right)^{-4} =$ __________

b. $\left(\frac{3}{-5}\right)^{-9} =$ __________

c. $\left(-\frac{8}{9}\right)^{-2} =$ __________

d. $\left(\frac{21}{4}\right)^{-5} =$ __________

3.5.3 Collaboration: Applying More Than One Exponent Law

1. Fernanda and Junior were asked to write an equivalent expression of $\left(\frac{2.5^5}{2.5^3}\right)^{-1}$ by applying multiple exponent laws. Each student first described their plan.

a. Complete each student's work based on the description of their plan.

Fernanda	Junior
I will first apply the **quotient of powers** exponent law. Then, I will apply the **power of a power** exponent law. Finally, I will re-write the decimal as a fraction and apply the **negative** exponent law.	I will first apply the **power of a quotient** exponent law. Then, I will apply the **quotient of powers** exponent law. Finally, I will re-write the decimal as a fraction and apply the **negative** exponent law.
$\left(\frac{2.5^5}{2.5^3}\right)^{-1}$ $= (2.5^2)^{-1}$ $= (2.5)^{-2}$ $= \left(\frac{5}{2}\right)^{-2}$ $=$	$\left(\frac{2.5^5}{2.5^3}\right)^{-1}$ $= \frac{(2.5)^{-5}}{(2.5)^{-3}}$ $=$ $=$ $=$

b. Explain which method you prefer.

c. Ask your partner which method they prefer. Summarize your partner's explanation.

3.5.4 Collaboration: Equivalent Expressions

1. Consider the expression $\frac{\left(\frac{1}{6}\right)^3}{6^{-5}} \cdot 6^4$.

   For each expression, work with your partner to determine if it is equivalent to $\frac{\left(\frac{1}{6}\right)^3}{6^{-5}} \cdot 6^4$. Then, justify your choice by explaining using laws of exponents or showing your work.

Expression	Is it Equivalent to $\frac{\left(\frac{1}{6}\right)^3}{6^{-5}} \cdot 6^4$?	Justify or Explain
a. $\frac{6^{-3}}{6^{-5}} \cdot 6^4$	○ Yes ○ No	
b. $6^{-8} \cdot 6^4$	○ Yes ○ No	
c. $\left(\frac{1}{6}\right)^{-2} \cdot 6^4$	○ Yes ○ No	
d. $\left(\frac{1}{6}\right)^{-8} \cdot 6^4$	○ Yes ○ No	
e. $\frac{\left(\frac{1}{6}\right)^{-3}}{\left(\frac{1}{6}\right)^5} \cdot \left(\frac{1}{6}\right)^{-4}$	○ Yes ○ No	
f. 6^6	○ Yes ○ No	

3.5.5 Your Turn!

1. Match each expression in Column A to an equivalent expression in Column B by drawing arrows.

Column A	Column B
$\left(-\frac{7}{8}\right)^{-2}$	$\left(\frac{7}{8}\right)^{2}$
$\left(-\frac{8}{7}\right)^{-2}$	$\left(-\frac{8}{7}\right)^{2}$
$\left(\frac{7}{8}\right)^{-2}$	$\left(\frac{8}{7}\right)^{2}$
$\left(\frac{8}{7}\right)^{-2}$	$\left(-\frac{7}{8}\right)^{2}$

2. Three expressions are shown. Circle the two expressions that are equivalent.

$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^{4}$ $\qquad$ $\dfrac{\left(\frac{5}{8}\right)^{-2}}{\left(\frac{5}{8}\right)^{-4}}$ $\qquad$ $\left(\left(\frac{5}{8}\right)^{-2}\right)^{4}$

3. Apply the laws of exponents to write an equivalent numerical expressions with one base and one positive exponent. Show your steps in the column on the left and identify the law of exponents used in each step in the column on the right.

$$\left[\left(\frac{2}{3}\right)^{5} \cdot \left(\frac{2}{3}\right)^{-2}\right]^{3}$$

Steps	Laws of Exponents

3.5.6 Wrap-Up: Ticket Out the Door

1. Rewrite each expression as an equivalent exponential expression with positive exponents.

   a. $\left(\frac{-13}{2}\right)^{-3} =$ ___________

   b. $\left(\frac{1}{5}\right)^{-8} =$ ___________

2. Generate an equivalent expression using the laws of exponents. Express your answer with positive exponents.

$$\left(\frac{-5}{4}\right)^{-7} \cdot \left(\frac{-5}{4}\right)^{-3}$$

Unit 3, Lesson 6: Evaluating Numerical Expressions with Integer Exponents

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Learning Target

☐ I can evaluate a numerical expression using the laws of exponents.